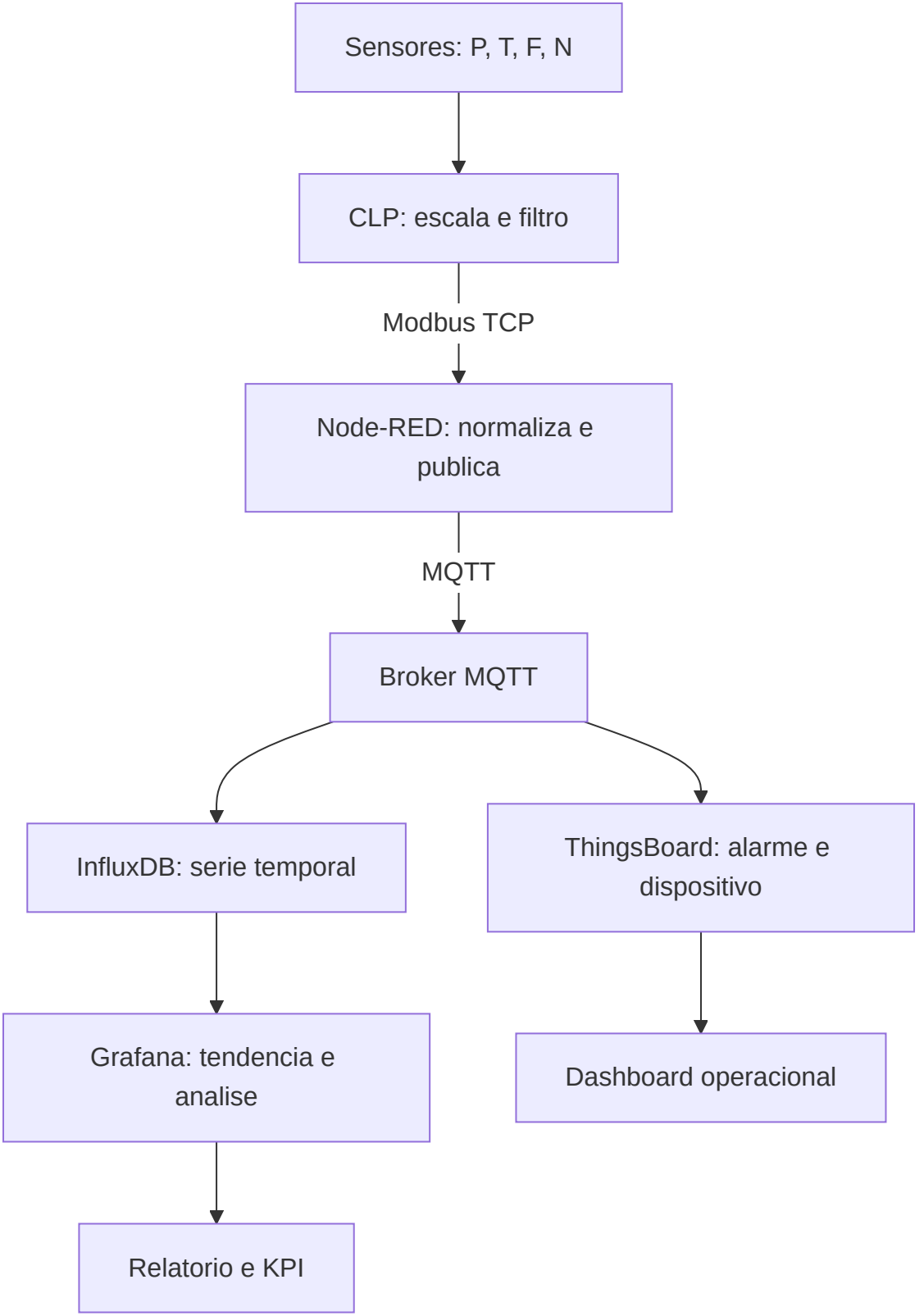
Sensores: P, T, F, N
CLP: escala e filtro
Modbus TCP
Node-RED: normaliza e publica
MQTT
Broker MQTT
InfluxDB: serie temporal
ThingsBoard: alarme e dispositivo
Grafana: tendencia e analise
Dashboard operacional
Relatorio e KPI